

within molecules is fundamental to chemistry. However, simulating these interactions accurately becomes increasingly complex with larger molecules. Quantum AI offers a path forward by leveraging its ability to model quantum systems efficiently. This allows chemists to design new materials, catalysts, and pharmaceuticals with previously unimaginable properties.

These are just a few examples of how Quantum AI is poised to revolutionise various fields. As quantum computing technology continues to evolve, we can expect even more groundbreaking applications to emerge, tackling problems that were once considered beyond the reach of classical computers.

Quantum neural networks

Quantum neural networks (QNNs) represent a fascinating intersection of artificial intelligence and quantum mechanics. They borrow the structure of classical artificial neural networks (ANNs) but leverage the power of qubits and quantum operations to tackle problems intractable for classical computers. They have the following characteristics:

Qubits vs bits. Classical ANNs rely on bits, which can be either 0 or 1. QNNs, however, utilise qubits, the quantum equivalent of bits. Qubits can exist in a superposition of states (0 and 1 simultaneously). This allows QNNs to explore a vast amount of data simultaneously, leading to potential speedups for specific problems.

Quantum gates. Classical ANNs use mathematical functions (e.g., sigmoid) as activation functions. QNNs employ quantum gates (e.g., Hadamard gate, Pauli-X gate) to manipulate qubits. These gates perform operations on qubits while preserving the superposition of states. Unlike classical activation functions, quantum gates are reversible, meaning

the operation can be undone.

Their features are:

Parallelism. Superposition allows QNNs to explore multiple possibilities at once. Imagine a maze—a classical computer would have to explore each path one by one, while a QNN can explore all paths simultaneously, potentially finding the exit much faster for certain problems.

Expressive power. Entanglement, another mind-bending principle of quantum mechanics, allows qubits to be linked. When entangled, the fates of these qubits are intertwined. This lets QNNs represent complex relationships between data points more efficiently compared to classical ANNs.

This is how they work:

Input layer. Similar to classical ANNs, QNNs have an input layer that receives data. This data is then encoded into qubits, representing the initial state of the network.

Hidden layers. QNNs consist of multiple hidden layers, each containing qubits. Quantum gates are applied sequentially to these qubits, manipulating their superpositions and entanglements. These operations transform the input data, allowing the network to learn patterns and features.

Output layer. The final layer of the QNN also contains qubits. Here, a measurement is performed, collapsing the superposition of states into a single classical output. This output could be a classification (e.g., identifying a handwritten digit), a prediction (e.g., forecasting stock prices), or some other relevant information.

QNNs are a nascent field with significant hurdles. Building and controlling large-scale quantum computers needed for powerful QNNs remains a challenge. Additionally, training QNNs is complex and requires specialised algorithms. Despite the challenges, QNNs hold

immense potential for applications in various domains.

The complexity of designing quantum algorithms that take advantage of the distinctive characteristics of qubits while also overcoming the constraints of classical computing presents a challenge for researchers and developers, as it necessitates proficiency in both quantum physics and artificial intelligence.

Quantum computing resources are presently constrained in terms of qubit count, coherence time, and gate fidelity; these limitations impede the efficacy and scalability of quantum AI algorithms. It is essential to scale quantum systems and enhance their hardware capabilities in order to fully exploit their potential.

Further progress in quantum hardware will result from ongoing research and development. These efforts will contribute to the fabrication of quantum processors that are more stable and capable of handling more complex quantum algorithms. Such processors will feature increased qubit counts, coherence periods, and gate fidelities.

Hybrid quantum-classical approaches are expected to gain prominence in the near future. These algorithms capitalise on the respective advantages of classical and quantum computing paradigms to tackle a diverse array of artificial intelligence tasks efficiently and effectively.

The commercialisation and adoption of quantum AI solutions will occur in tandem with the maturation of quantum computing technologies. This will bring about significant transformations in various sectors, including healthcare, finance, logistics, and cybersecurity. **EFY**

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